AWS Capstone Project Submission

Student Name: Sumaiya Sara
Project: AWS Cloud Architecting Capstone Project
Status: Completed Successfully

Project Overview

This project was completed using AWS cloud services to build a secure, highly available web application architecture. The application was deployed using EC2 instances in private subnets, an Application Load Balancer, Auto Scaling Group, RDS MySQL database, and AWS Secrets Manager.

AWS Services Used

- Amazon EC2
- Amazon RDS MySQL
- Application Load Balancer (ALB)
- Auto Scaling Group (ASG)
- AWS Secrets Manager
- Amazon VPC
- Security Groups
- Private and Public Subnets

Completed Tasks

- Created MySQL RDS database in private subnets
- Configured AWS Secrets Manager for secure credentials storage
- Created Application Load Balancer in public subnets
- Configured Auto Scaling Group across two Availability Zones
- Launched EC2 instances using Project Launch Template
- Configured security groups between ALB, EC2, and RDS
- Imported SQL country dataset into RDS database
- Successfully tested application connectivity
- Verified application accessibility through ALB DNS
- Received full marks on project submission

Architecture Flow

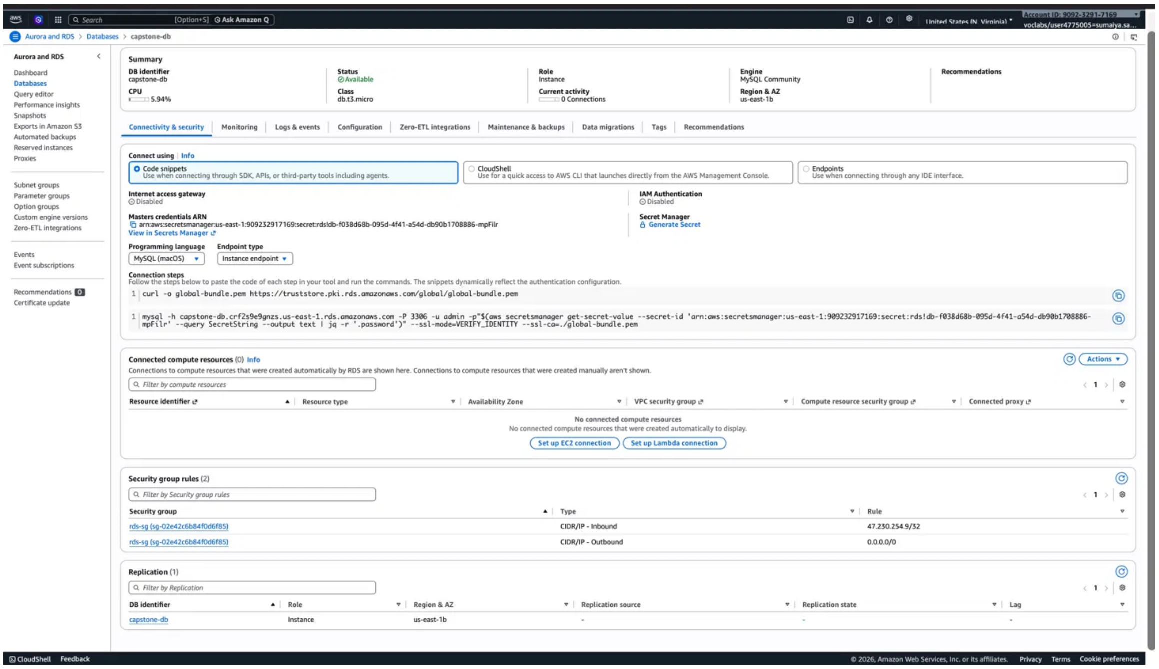
Internet → Application Load Balancer → EC2 Instances (Private Subnets) → RDS MySQL Database → Query Results Returned to User

Submission Notes

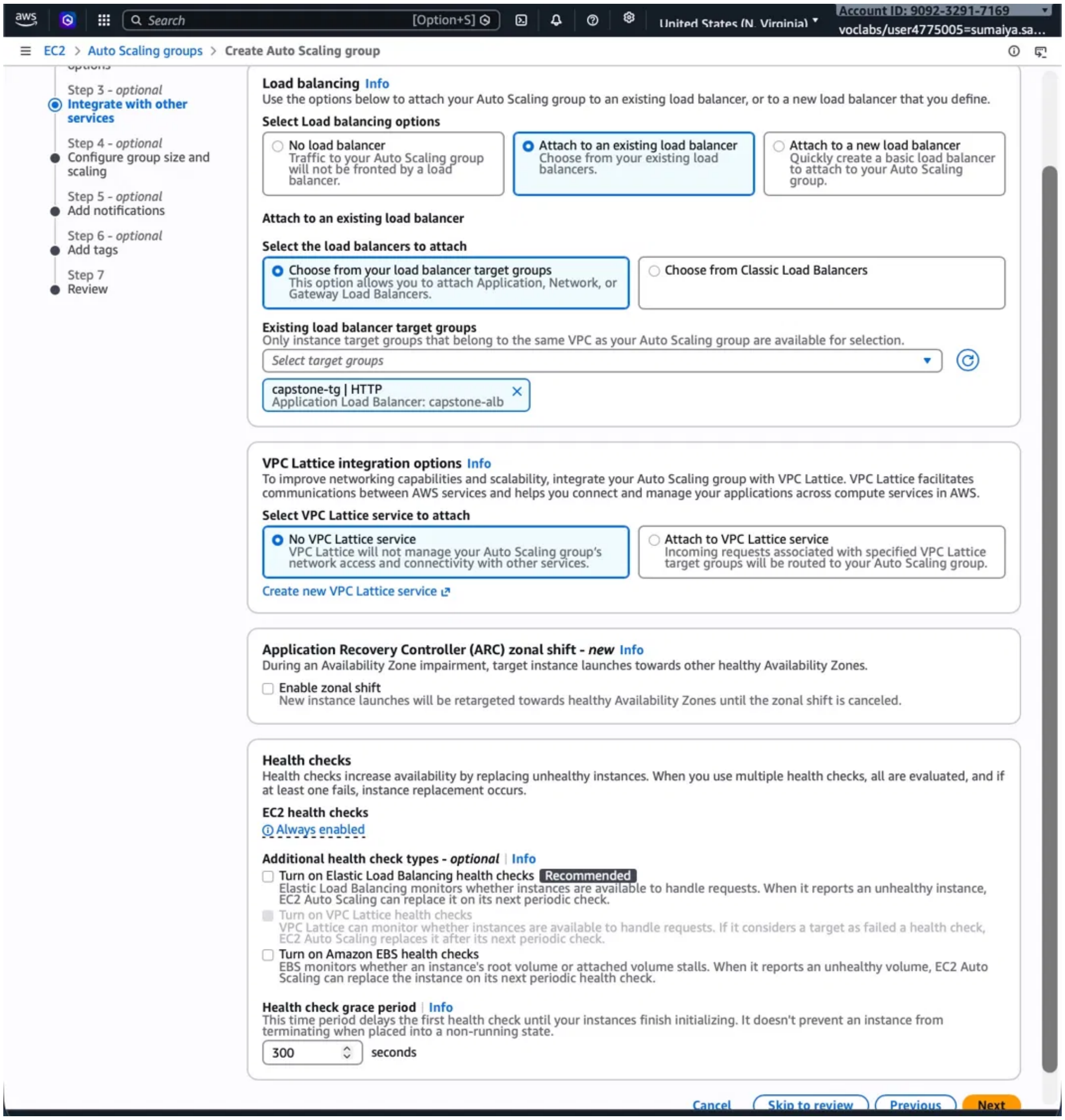
The project was fully completed and successfully submitted through the AWS Academy lab environment. The grading report confirmed that the application was reachable, the database was private, Secrets Manager was used, and EC2 instances were launched correctly in the Auto Scaling Group.

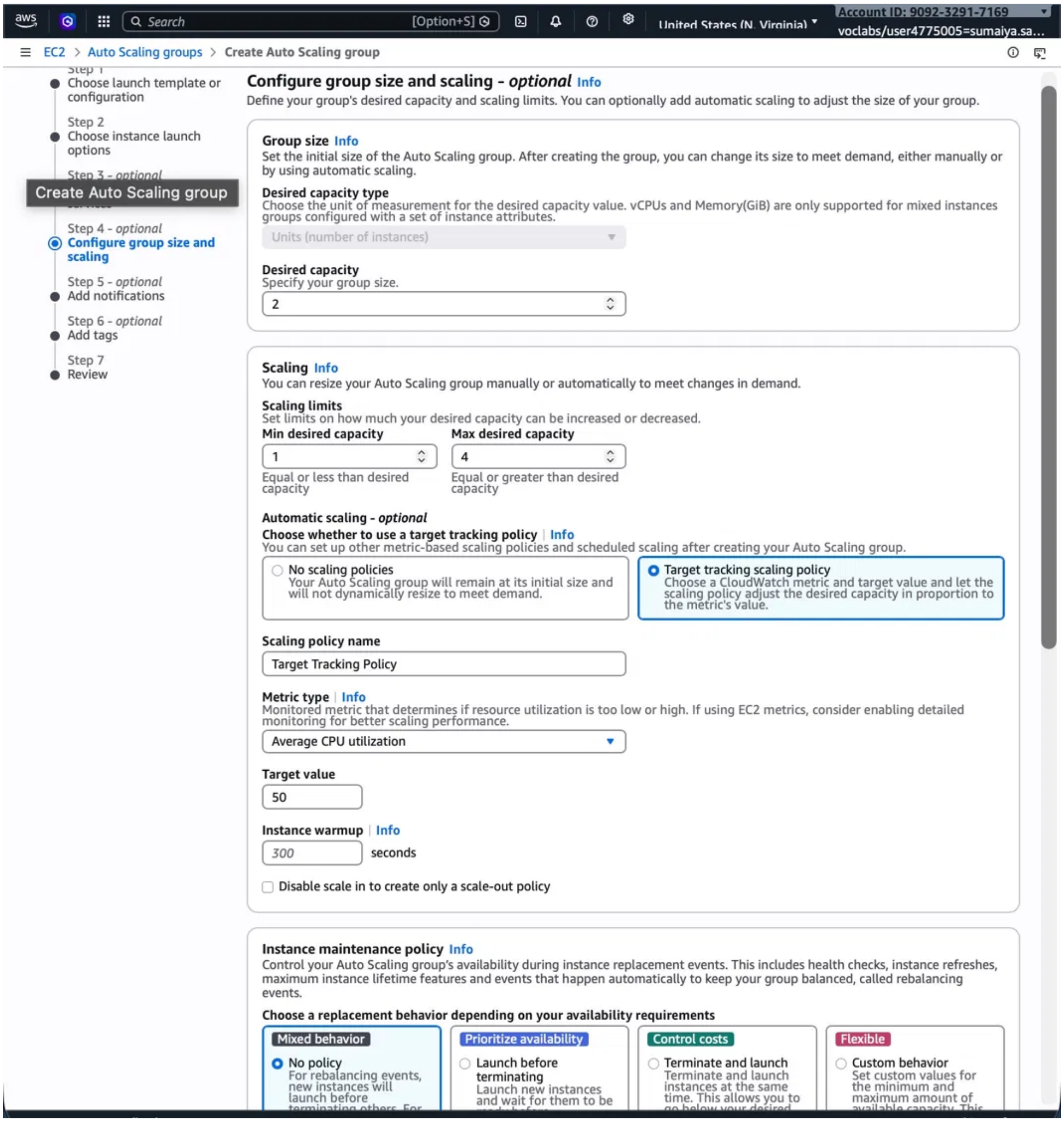
Recommended Screenshot Sections

1. RDS Database Available

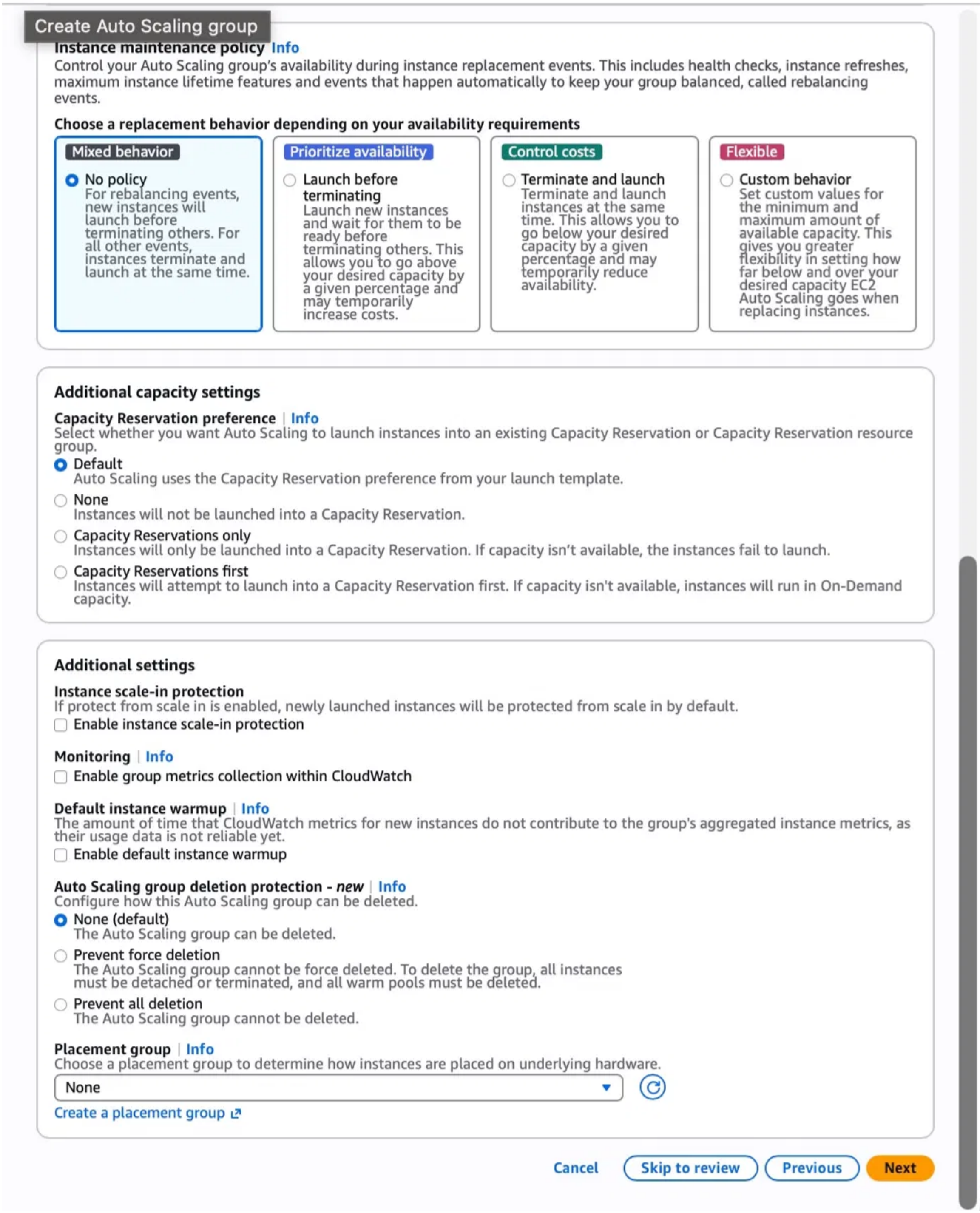


2. Load Balancer Created

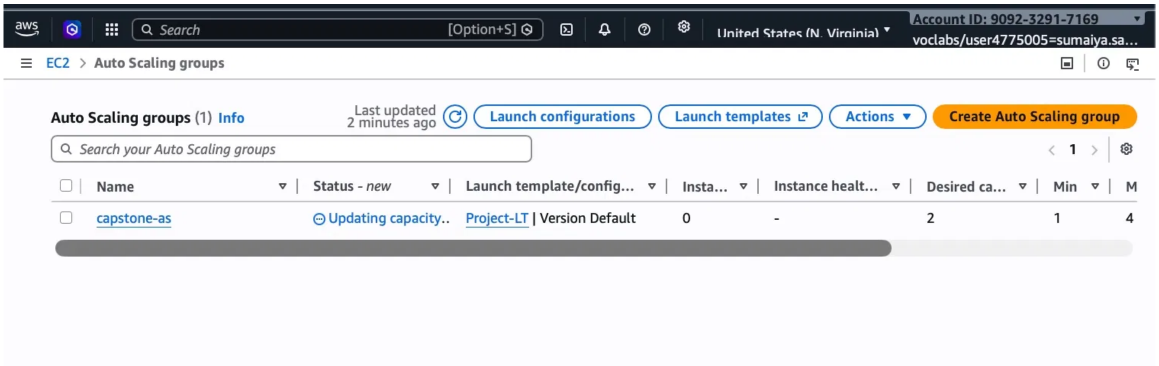

Search
[Option+S]
United States (N. Virginia)
Account ID: 9092-3291-7169
voclabs/user4775005=sumaiya.sa...
EC2 > Auto Scaling groups > Create Auto Scaling group
Step 3 - optional
Integrate with other services
Step 4 - optional
Configure group size and scaling
Step 5 - optional
Add notifications
Step 6 - optional
Add tags
Step 7
Review
Load balancing Info
Use the options below to attach your Auto Scaling group to an existing load balancer, or to a new load balancer that you define.
Select Load balancing options
No load balancer
Traffic to your Auto Scaling group will not be fronted by a load balancer.
Attach to an existing load balancer
Choose from your existing load balancers.
Attach to a new load balancer
Quickly create a basic load balancer to attach to your Auto Scaling group.
Attach to an existing load balancer
Select the load balancers to attach
Choose from your load balancer target groups
This option allows you to attach Application, Network, or Gateway Load Balancers.
Choose from Classic Load Balancers
Existing load balancer target groups
Only instance target groups that belong to the same VPC as your Auto Scaling group are available for selection.
Select target groups
capstone-tg | HTTP
Application Load Balancer: capstone-alb
VPC Lattice integration options Info
To improve networking capabilities and scalability, integrate your Auto Scaling group with VPC Lattice. VPC Lattice facilitates communications between AWS services and helps you connect and manage your applications across compute services in AWS.
Select VPC Lattice service to attach
No VPC Lattice service
VPC Lattice will not manage your Auto Scaling group's network access and connectivity with other services.
Attach to VPC Lattice service
Incoming requests associated with specified VPC Lattice target groups will be routed to your Auto Scaling group.
Create new VPC Lattice service
Application Recovery Controller (ARC) zonal shift - new Info
During an Availability Zone impairment, target instance launches towards other healthy Availability Zones.
Enable zonal shift
New instance launches will be retargeted towards healthy Availability Zones until the zonal shift is canceled.
Health checks
Health checks increase availability by replacing unhealthy instances. When you use multiple health checks, all are evaluated, and if at least one fails, instance replacement occurs.
EC2 health checks
Always enabled
Additional health check types - optional Info
Turn on Elastic Load Balancing health checks Recommended
Elastic Load Balancing monitors whether instances are available to handle requests. When it reports an unhealthy instance, EC2 Auto Scaling can replace it on its next periodic check.
Turn on VPC Lattice health checks
VPC Lattice can monitor whether instances are available to handle requests. If it considers a target as failed a health check, EC2 Auto Scaling replaces it after its next periodic check.
Turn on Amazon EBS health checks
EBS monitors whether an instance's root volume or attached volume stalls. When it reports an unhealthy volume, EC2 Auto Scaling can replace the instance on its next periodic health check.
Health check grace period Info
This time period delays the first health check until your instances finish initializing. It doesn't prevent an instance from terminating when placed into a non-running state.
300
seconds
Cancel
Skip to review
Previous
Next



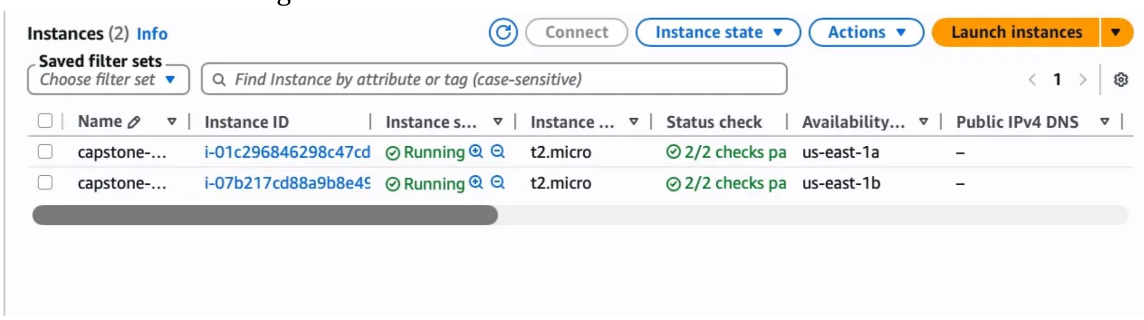
Search
[Option+S]
United States (N. Virginia)
Account ID: 9092-3291-7169
voclabs/user4775005=sumaiya.sa...
EC2 > Auto Scaling groups > Create Auto Scaling group
Step 1
Choose launch template or configuration
Step 2
Choose instance launch options
Step 3 - optional
Create Auto Scaling group
Step 4 - optional
Configure group size and scaling
Step 5 - optional
Add notifications
Step 6 - optional
Add tags
Step 7
Review
Configure group size and scaling - optional Info
Define your group's desired capacity and scaling limits. You can optionally add automatic scaling to adjust the size of your group.
Group size Info
Set the initial size of the Auto Scaling group. After creating the group, you can change its size to meet demand, either manually or by using automatic scaling.
Desired capacity type
Choose the unit of measurement for the desired capacity value. vCPUs and Memory(GiB) are only supported for mixed instances groups configured with a set of instance attributes.
Units (number of instances)
Desired capacity
Specify your group size.
2
Scaling Info
You can resize your Auto Scaling group manually or automatically to meet changes in demand.
Scaling limits
Set limits on how much your desired capacity can be increased or decreased.
Min desired capacity
1
Equal or less than desired capacity
Max desired capacity
4
Equal or greater than desired capacity
Automatic scaling - optional
Choose whether to use a target tracking policy Info
You can set up other metric-based scaling policies and scheduled scaling after creating your Auto Scaling group.
No scaling policies
Your Auto Scaling group will remain at its initial size and will not dynamically resize to meet demand.
Target tracking scaling policy
Choose a CloudWatch metric and target value and let the scaling policy adjust the desired capacity in proportion to the metric's value.
Scaling policy name
Target Tracking Policy
Metric type Info
Monitored metric that determines if resource utilization is too low or high. If using EC2 metrics, consider enabling detailed monitoring for better scaling performance.
Average CPU utilization
Target value
50
Instance warmup Info
300
seconds
Disable scale in to create only a scale-out policy
Instance maintenance policy Info
Control your Auto Scaling group's availability during instance replacement events. This includes health checks, instance refreshes, maximum instance lifetime features and events that happen automatically to keep your group balanced, called rebalancing events.
Choose a replacement behavior depending on your availability requirements
Mixed behavior
No policy
For rebalancing events, new instances will launch before
Prioritize availability
Launch before terminating
Launch new instances and wait for them to be
Control costs
Terminate and launch
Terminate and launch instances at the same time. This allows you to
Flexible
Custom behavior
Set custom values for the minimum and maximum amount of



3. Auto Scaling Group Running



4. EC2 Instances Running



5. Working Website Homepage

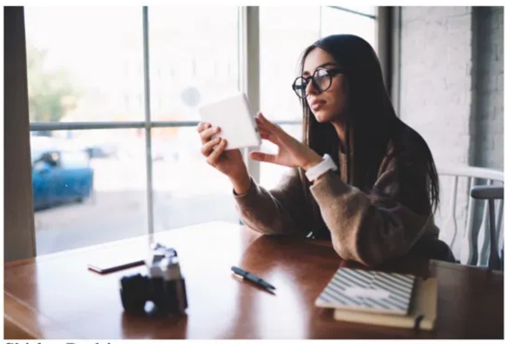
Example Social Research Organization

About Us Contact Us Query

Welcome to our data query site. You can get data from countries all over the world to use in your research.

We provide data for a variety of areas including basic demographics and development statistics.

About Us



Shirley Rodriguez

Our site got started when Shirley Rodriguez found that she was frequently looking up data from a variety of databases.
Shirley decided to start sharing some of this data with other social researchers.

Contact Us

6. Database Query Results

Pick another query

Country Name	Population	Urban Population
Afghanistan	26697430	5771984
Albania	3071856	1280964
Algeria	30533827	18259229
American Samoa	57625	51171
Andorra	64634	59722
Angola	13926373	6823923
Antigua and Barbuda	77656	24928
Argentina	36930709	33274569
Armenia	3076098	2002540
Aruba	90271	42157
Australia	19153000	16701416
Austria	8011566	5271610
Azerbaijan	8048600	4120883
Bahamas, The	297651	244074
Bahrain	638193	564163
Bangladesh	129592275	30583777
Barbados	267511	97106
Belarus	10005000	6993495
Belgium	10251250	9953964
Belize	249800	119404
Benin	6517810	2496321
Bermuda	62100	62100
Bhutan	571262	145101
Bolivia	8307248	5133879
Bosnia and Herzegovina	3693698	1595678
Botswana	1757925	935216
Brazil	174425387	141633414
Brunei Darussalam	327036	232523
Bulgaria	8170172	5629249
Burkina Faso	12294012	2040806
Burundi	6374347	529071
Cambodia	12446949	2103534
Cameroon	15678269	7823456
Canada	30769700	24461912
Cape Verde	437238	233485

7. Submission Report with Full Marks

Capstone Project

Due No Due Date **Points** 35 **Submitting** an external tool

AWS Used $0 of $60 01:11 Start Lab End Lab AWS Details Details Reset

Submit Submission Report Grades

submission is recorded for this lab.

Ending your session

Reminder: This is a long-lived lab environment. Data is retained until either you use the allocated budget or the course end date is reached (whichever occurs first).

To preserve your budget when you are finished for the day or when you are finished actively working on the assignment for the time being, do the following:

16. At the top of this page, choose **End Lab**, and then choose Yes to confirm that you want to end the lab.

A message appears: *Ended AWS Lab Successfully*

Note: Choosing **End lab** in this lab environment will not delete the resources that you have created. They will still be there the next time you choose **Start Lab** (for example, on another day).

Total score	35/35
[Task 1] Load Balancer is created	5/5
[Task 2] Application is reachable	5/5
[Task 3] Application Server is created in Private Subnets	5/5
[Task 4] RDS MySQL Database is created	5/5
[Task 5] Database uses Secrets Manager	5/5
[Task 6] Database is not public	5/5
[Task 7] EC2 Instances are launched in Auto Scaling group	5/5